

The Acute Abdomen: A Case-based Survival Guide to What the Surgeon Wants to Know

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See the slide presentation [here](#).

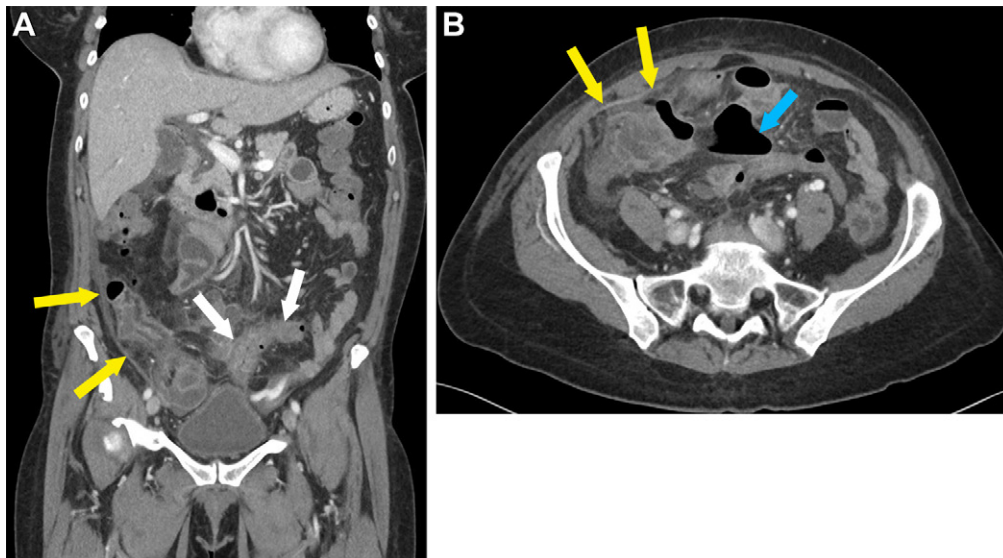
Acute abdominal pain is a leading cause of emergency department visits, accounting for up to 10% of visits annually. Both benign and emergent abdominal pathologic conditions can manifest with similar history and physical examination findings. Timely diagnoses are crucial for making appropriate management decisions, and imaging is routinely used in evaluating for underlying causes and potential complications that require surgical intervention. The radiologist, therefore, serves a vital role in not only making the diagnosis but also providing key findings to the clinical and surgical teams that can influence and improve patient outcomes.

A review of every cause and manifestation of the acute surgical abdomen is impractical. However, the principles that influence surgical management can be broadly defined to five

categories: obstruction, ischemia, perforation, active bleeding, and aggressive infections. By understanding how these categories apply to common manifestations, radiologists can construct their reports with pertinent findings that aid in urgent management decisions. For example, reporting distant free intrabdominal air and peritonitis in acute diverticulitis may influence the surgeon to proceed with urgent operative management (Fig 1).

Conversely, reviewing the classifications used by surgeons can help the radiologist understand which negative findings to include. For example, reporting the absence of bowel wall ischemia, pneumatosis, and perforation in a small bowel obstruction may encourage conservative therapy and prevent unnecessary trips to the operating room (Fig 2).

Figure 1. Acute sigmoid diverticulitis in a 65-year-old woman presenting with 3 weeks of lower abdominal pain. Coronal (A) and axial (B) CT images show acute sigmoid diverticulitis (white arrows in A). A 4.5-cm interloop abscess separate from the sigmoid colon (blue arrow in B) is present in the central mesentery due to perforation. Additional thickening and hyperenhancement of the peritoneum in the right lower quadrant (yellow arrows) is present, consistent with distant peritonitis. This patient required surgical management.





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TEACHING POINTS

- The acute abdomen is a commonly encountered scenario in the emergent setting.
- Five major categories of findings can influence emergent surgical intervention: obstruction, ischemia, perforation, infections, and bleeding.
- By recognizing specific imaging findings, radiologists can tailor their reports to provide the most relevant information and guide surgical decision making.

The slide presentation reviews the most common causes of the acute surgical abdomen through a case-based format. Each section reviews potential complications and provides reporting recommendations targeted toward radiology residents who are taking call. By the end of the presentation, the viewer should feel comfortable providing a focused summary of the pertinent findings that are important to the surgeon.

In summary, the acute abdomen is a commonly encountered manifestation in the emergency setting, and imaging plays a vital role in establishing a timely diagnosis and management. Knowledge of the general principles influencing surgical management can allow radiologists to add value to their reports and improve patient outcomes.

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Suggested Readings

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Figure 2. Acute obstruction in a 20-year-old man presenting with abdominal pain, nausea, and vomiting for 1 day. Axial (A) and coronal (B) CT images show dilated loops of small bowel with a focal transition point into decompressed bowel in the central abdomen (white arrow in B), consistent with an acute obstruction. Normal enhancement of the bowel wall (yellow arrows) without evidence of vascular compromise or perforation is present. The patient was conservatively managed with decompression and was discharged without surgical intervention.

